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Assignment Title:	Conducting a Systematic Literature Review (SLR) in Computer Science		
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Q1. Formulating Research Questions (RQs)

Answer:

A possible hypothetical research project is Large Language Models (LLMs) For Automated Code Generation. The Systematic Literature Review (SLR) aims to define the objectives of the SLR by determining the relevant search keywords based on the components of the framework of PICOC. PICOC is a relatively new method that leads the way for the researcher to think about the essential elements of a research question. It has been used in medical and social science related research works, but in software engineering related research works, it can be effectively used. The PICOC elements in this research are determined based on the framework that can be seen in Table 1 which will be the basis of forming research questions.

PICOC Element	What it means for this topic
Population	Software developers, programmers, and software development activities.
Intervention	Large Language Models (LLMs) used for generating source code.
Comparison	Traditional programming methods and older automated code generation techniques.
Outcome	Code of correctness, quality, reliability, and maintainability.
Context	Software development environments and real-world programming projects.

Table-1: Meaning For PICOC keywords

Main Research Question (MRQ)

MRQ:

Can large language models create valid and sustainable source code more effectively than code creation methods?

Sub-Research Questions (SRQs)

SRQ1:

What are the typical metrics, benchmarks, and methods used to assess the quality and correctness of the LLM-generated code?

SRQ2:

What are some problems or weaknesses shared by LLM-based automatic code generation in real software development?

Q2. Defining Search Strings

Answer:

Large Language Models (LLMs) for Automated Code Generation an effective search strategy requires identifying the main concepts of the research topic, their synonyms, and appropriate Boolean operators. The following steps define the keywords and construct a general search string for the proposed SLR.

Step 1: Keywords and Synonyms per PICOC Element

PICOC Element	Keywords	Synonyms
Population	Software Developers	Programmers, Software Engineers, Developers.
Intervention	Large Language Models	LLMs, Generative AI, Foundation Models, AI-Assisted Code Generation.
Comparison	Traditional Code Generation Methods	Conventional Methods, Rule-Based Methods, Template-Based Methods, Manual Coding.
Outcome	Code Quality	Correctness, Maintainability, Reliability, Functional Correctness, Software Quality.
Context	Automated Code Generation	Automatic Code Generation, Code Synthesis, Program Synthesis, Software Development.

TABLE-2: PICOC Synonyms Findings

Step 2: General Boolean Search String

The keywords and synonyms identified in Step 1 are combined using Boolean operators such as AND and OR. The OR operator broadens the search by including synonyms, while AND connects the major concepts of the research topic.

General Search String:

("Large Language Model" OR "LLM" OR "Generative AI" OR "Claude" OR "Codex" OR "Transformer Model") AND ("Code Generation" OR "Automated Programming" OR "Program Synthesis" OR "Code Completion") AND ("Software Developer" OR "Programmer" OR "Software Engineer") AND ("Traditional Code Generation" OR "Conventional Method" OR "Rule-Based Method" OR "Template-Based Method" OR "Manual Coding") AND ("Code Correctness" OR "Code Quality" OR "Software Development" OR "Software Engineering") AND "Accuracy" OR "Reliability" OR "Maintainability" OR "Evaluation".

Q3. Search Strategy & Selection Process

Answer:

a. Database Selection

An SLR should look in several academic databases for a comprehensive and trustworthy set of results of relevant studies.

Database	Justification
IEEE Xplore	It includes access to scientific and technical publications such as: Journal articles, conference Proceedings, and standards. It has a high relevance to Also, there are studies in software engineering, artificial intelligence and robotics. In addition, there are software engineering, AI and robotics studies. automated code generation
ACM Digital Library	It is a primary source for journals, conference proceedings and other computing-related research. It is particularly suitable for locating research on programming languages, software engineering, and code generation.
Scopus	It is a primary source for journals, conference proceedings and other computing (35000+) reviewed journals. All proceedings, and other

	publications in computer science and information technology. Its citation and analytical features can help identify relevant studies and trace influential research.
arXiv	It is an important source for recent computer science and machine learning research. In science and technology, conference proceedings are available. However, LLM technology is also evolving quickly, and arXiv can be used to identify. Emerging Studies and preprints that are not yet published in the traditional journal's publication venues.

TABLE-3: Database Justification

b. Inclusion and Exclusion Criteria

Inclusion Criteria

- The studies published in English between 2020 and 2026, including peer-reviewed journal articles. Articles, conference papers and relevant preprints.
- Using code generation, code synthesis, or code analysis (such as code search) to generate code. Methods that study automated code generation, code synthesis, or code analysis for the purpose of generating code. Completeness, correctness, quality, maintainability of code.

Exclusion Criteria

- Processing cannot adequately assess their potential and limitations for more specialized tasks like content creation, translation, and face and object recognition. Research that investigates only non-specific utilization of LLMs, like natural language generation or image generation instead of source code generation.
- Non-English publications, duplicate studies and articles with limited full-text information about assessing their relevance and findings.